

HW 2

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Q-1

- 1) $h(n)$ = number of positions where n differs from the goal.
The steps heuristic counts the number of positions where the current word differs from the goal. Since each action changes exactly one position, at least one step is required per mismatched position. Therefore the heuristic never overestimates the true remaining cost.
- 2) $h(n)$ = sum of SCRABBLE_VALUES[goal[i]] for each position i where the current word differs from the goal. For each mismatched position, the goal letter must eventually be placed there, costing exactly its scrabble tile value. Since each action changes only one position, these costs are paid separately, so the sum is a lower bound and never overestimates.
- 3) $h(n)$ = number of mismatched positions + rarity(goal). Any solution must take at least as many steps as there are mismatched positions, and the final step must enter the goal word, costing 1 + rarity(goal). The remaining steps each cost at least 1, so the total remaining cost is at least count + rarity(goal), meaning the heuristic never overestimates.

Q-2

16 paths have been expanded and 73 paths remain in the frontier

773 paths have been expanded and 119 paths remain in the frontier

UCS cost: 5.00

dc(why,ask,steps) cost:5.00; time:0.547; solution:why wha aha ahs ass ask; deltas:[-2, -1, -1, 0.0, 0.0]

13 paths have been expanded and 70 paths remain in the frontier

756 paths have been expanded and 161 paths remain in the frontier

UCS cost: 9.00

dc(why,ask,scrabble) cost:9.00; time:1.008; solution:why wha aha ahs ass ask; deltas:[-2, -1, -1, 0.0,0.0]

634 paths have been expanded and 213 paths remain in the frontier

808 paths have been expanded and 89 paths remain in the frontier

UCS cost: 38.86

dc(why,ask,frequency) cost:38.86; time:1.761; solution:why way was has his ais ass ask; deltas:[-31.623, -28.685, -27.217, -25.559, -23.591, -9.704, 0.0]

6 paths have been expanded and 54 paths remain in the frontier

691 paths have been expanded and 1409 paths remain in the frontier

UCS cost: 4.00

dc(bear,duck,steps) cost:4.00; time:1.084; solution:bear beak beck buck duck; deltas:[0.0, 0.0, 0.0, 0.0]

6 paths have been expanded and 54 paths remain in the frontier

2952 paths have been expanded and 535 paths remain in the frontier

UCS cost: 11.00

dc(bear,duck,scrabble) cost:11.00; time:6.085; solution:bear beak beck buck duck; deltas:[0.0, 0.0, 0.0, 0.0]

302 paths have been expanded and 806 paths remain in the frontier

1084 paths have been expanded and 1528 paths remain in the frontier

UCS cost: 38.24

dc(bear,duck,frequency) cost:38.24; time:2.392; solution:bear beak beck buck duck; deltas:[-27.011, -15.546, -7.034, 0.0]

Word pair	Cost	UCS cost	UCS nodes	A* cost	A* nodes	Reduction
WHY→ASK	Steps	5.00	773	5.00	16	97.93%
WHY→ASK	Scrabble	9.00	756	9.00	13	98.28%
WHY→ASK	Frequency	38.86	808	38.86	634	21.53%
BEAR→DUCK	Steps	4.00	691	4.00	6	99.13%
BEAR→DUCK	Scrabble	11.00	2952	11.00	6	99.80%
BEAR→DUCK	Frequency	38.24	1084	38.24	302	72.14%

- 1) Yes, it did
- 2) For both WHY→ASK and BEAR→DUCK, A* produced the largest percentage reduction in expanded nodes under the Scrabble cost measure, with 98.28% and 99.80% reductions respectively.
- 3) For both WHY→ASK and BEAR→DUCK, the heuristic was least informative under the Frequency cost measure, with reductions of only 21.53% and 72.14% respectively.
- 4) The frequency heuristic is less informative than steps and scrabble because it only accounts for the number of mismatched positions and the rarity of the goal word. It ignores the rarity of all intermediate words along the path, which can be very high. This means the heuristic significantly underestimates the true remaining cost, resulting in less node reduction compared to the other cost measures.

Q-3

Steps: "Under the steps cost measure, the optimal cost from W1 to W2 always equals the optimal cost from W2 to W1. Since every action costs exactly 1 regardless of which letter is being replaced, reversing the path produces the same total cost."

Scrabble: "Under the scrabble cost measure, the optimal cost is not always symmetric. The cost of an action depends on the value of the replacement letter, which differs depending on direction. For example, pig→dig costs 2 (replacing p with d) while dig→pig costs 3 (replacing d with p)."

Frequency: "Under the frequency cost measure, the optimal cost is not always symmetric. The cost of entering a word is $1 + \text{rarity}(\text{word})$, so the path cost depends on which intermediate words are visited and their rarities. For example, dog→cat costs 23.02 while cat→dog costs 21.96.

Q-4

Model used: ChatGPT (GPT-4)

Proposed heuristic: $hR(n) = h0(n) \times (1 + r_{\min})$ where $r_{\min}$ is the minimum rarity across all dictionary words.

Admissibility check: The heuristic is technically admissible since every action costs at least $1 + r_{\min}$. However, since some words in the dictionary have $\text{rarity} = 0$, $r_{\min} = 0$, which means $hR(n) = h0(n) \times 1 = h0(n)$. The heuristic collapses to just counting mismatched positions and is no more informative than $h0$.

Fix: A better heuristic is $h(n) = \text{count} + \text{rarity}(\text{goal})$, where count is the number of mismatched positions. This uses the rarity of the specific goal word rather than the global minimum, making it more informative while remaining admissible.

Test results: Using our heuristic on the frequency problems from Q2, WHY→ASK expanded 634 nodes and BEAR→DUCK expanded 302 nodes with A*. ChatGPT's heuristic would expand the same number of nodes as the steps heuristic since it collapses to $h0$.

AI used for neatly writing the answers and draw me a table

